Saturday

Train a student prediction model

Saturday | Sunday | Deep Learning Weekend

PyTorch foundations

Slides 2-33

Explore tensors, shapes, linear layers and activations. Convert logits with sigmoid and trace how loss and autograd guide weight updates.

Prepare the student dataset

Slides 34-44

Load the Module 3 CSV with pandas. Select study hours and attendance as features and passed as the target. Make a stratified split and convert the tables to tensors.

Train and validate the model

Slides 45-57

Set a majority-class baseline. Use TensorDataset and DataLoader to build batches, then run the forward pass, loss, backpropagation and optimizer update.

Improve and select a model

Slides 58-74

Read training and validation loss curves. Explore optimizers, regularization, early stopping and checkpoints. Compare learning rates before evaluating on the test set.

Save the model and predict

Slides 75-80

Save the model weights, scaling settings and held-out row indices. Use the saved preprocessing to predict Pass or Fail for a new student.

Practice checklist

- [ ] Prepare the pandas feature and target tables
- [ ] Explain the four steps of a training update
- [ ] Run a prediction for a new student

Sunday

Explore vision, language, and inference

Image tensors and CNNs

Slides 2-27

Prepare image tensors and inspect their shapes. Use convolution, activation and pooling to train a small image classifier.

Transfer learning and evaluation

Slides 28-41

Use a pretrained vision model. Freeze its feature extractor or fine-tune it, then inspect class-wise errors and consider compute needs.

Tokens, embeddings, and attention

Slides 42-55

Turn text into tokens, map them to embeddings, and add positional information. Follow how attention combines information across tokens.

Transformers and text inference

Slides 56-69

Trace next-token prediction in a Transformer. Try pretrained text examples and practice saving, loading, and running inference.

Student project and serving

Slides 70-80

Use pandas to review prediction errors on Saturday’s saved test rows. Run the student model through Streamlit or serve it with FastAPI.

Practice checklist

- [ ] Explain what convolution and pooling do
- [ ] Trace the path from tokens to predictions
- [ ] Review errors on the saved student test rows